

Notes: Tetlock (JF, 2007)

Giving Content to Investor Sentiment: The Role of Media in the Stock Market

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1 Big picture

- **Question.** Does the *content* of financial media forecast broad stock market activity, or does it merely echo information already in prices?
- **Setting.** The paper studies the Wall Street Journal’s “Abreast of the Market” column from 1984 to 1999 and links its daily tone to U.S. stock returns and trading volume.
- **Core challenge.** A negative newspaper column could proxy for several very different things: investor sentiment, risk aversion, trading-cost shocks, genuine news about fundamentals, stale information already priced in, or simply irrelevant noise.
- **Main conceptual contribution.** The paper moves beyond headline events and studies a quantitative measure of *media tone*. It asks whether a systematic text-based measure of pessimism behaves like a proxy for sentiment or liquidity trading.
- **Main hypotheses.**
 - **Sentiment / noise-trader view:** high pessimism predicts temporary downward price pressure followed by reversal; unusually high or low pessimism predicts high volume.
 - **Fundamental-news view:** high pessimism predicts persistent negative returns, not reversal.
 - **Stale-information or sideshow view:** pessimism should not predict future market activity.
- **Main empirical design.** The author converts daily newspaper text into a pessimism factor using General Inquirer word categories and then estimates daily predictive regressions / VAR-style equations for Dow Jones returns, NYSE volume, and the Fama-French SMB factor.
- **Main baseline result.** A one-standard-deviation increase in media pessimism predicts about an **8.1 basis point** decline in next-day Dow Jones returns, followed by a **6.8 basis point** reversal over the next four trading days.
- **Key volume result.** The *absolute value* of pessimism predicts higher next-day volume. A one-standard-deviation increase in absolute pessimism raises detrended NYSE log volume by a bit more than **1%**, consistent with disagreement or noninformational trading.
- **Key small-stock result.** Pessimism predicts more persistent negative returns for the Fama-French **SMB** factor. The cumulative effect over the following week is about **-4.6 to -5.9 basis points**, and remains strong even when the sentiment measure is lagged an extra day.
- **Main credibility result.** The findings survive multiple checks: alternative sentiment proxies (“Negative” and “Weak” words), delayed return windows that begin at **10 a.m.** the next day, after-morning volume, subperiod analysis, and semiparametric lowess fits.
- **Main interpretation.** The evidence is most consistent with media tone proxying for investor sentiment, liquidity demand, or noise-trader beliefs. It is not consistent with media tone being a proxy for new fundamental information or lower risk.

- **Economic takeaway.** Financial media can help measure sentiment in real time, and sentiment appears to move prices temporarily—especially for small stocks—before those price effects reverse.

2 Data and measurement (key objects)

2.1 Why the WSJ “Abreast of the Market” column is the right laboratory

- The paper studies a daily Wall Street Journal column focused specifically on the stock market.
- Tetlock argues this source is attractive for three reasons:
 - the WSJ had the largest circulation of any U.S. daily financial publication (over 2 million readers);
 - Dow Jones Newswires reached over 325,000 finance and investment professionals;
 - electronic text for this column is available over a long horizon.
- The column is written around the market close and often reads like a post-mortem of the previous trading day.

Interpretation. The column is both salient and market-focused, making it a plausible object through which sentiment may be reflected or amplified.

2.2 Sample period and basic market data

- Main sample: roughly **January 1984 to September 17, 1999**.
- Effective regression sample: about **3,709 daily observations**.
- Each year contributes only about 250 observations because markets close on weekends and holidays.
- Return data for the Dow Jones Industrial Average come from WRDS / Dow Jones data.
- Detrended daily NYSE volume comes from the NYSE database.
- Small-stock returns use the daily Fama-French **SMB** factor.
- For timing-robustness checks, the paper also uses intraday Dow Jones index levels to build a return window starting at **10 a.m.** the day after the article is released.

2.3 From text to numbers: the General Inquirer categories

- For each day, the author counts words in **77 General Inquirer (GI)** categories from the Harvard IV-4 psychosocial dictionary.

- The GI does not interpret full sentences; it counts words that belong to categories such as:
 - **Negative**: words associated with a negative outlook,
 - **Weak**: words implying weakness,
 - **Fail**: words indicating goals are not achieved,
 - **Fall**: words related to falling movement,
 - and others such as **Positive**.
- Because writing style varies systematically by weekday, all GI series are **demeaned by day of the week** before factor extraction.

Interpretation. The paper intentionally uses a simple and reproducible text procedure rather than hand-coding tone.

2.4 Constructing the pessimism factor

The first media factor is obtained from a principal-components analysis of the daily GI-category covariance matrix. Let $w_t \in \mathbb{R}^{77}$ be the vector of demeaned GI word counts for day t . The paper constructs a one-factor summary

$$\text{BdNws}_t = a'w_t, \quad (1)$$

where a is the first principal-component loading vector.

Important implementation details:

- Loadings are estimated using only **year** $t - 1$ data and then applied to year t data.
- This avoids obvious look-ahead bias.
- The first factor is stable across years: the average pairwise correlation across yearly factors is about **0.96**, the average squared correlation is about **0.91**, and even the minimum pairwise correlation is about **0.80**.
- The average first eigenvalue is about **6.72**, so the first factor explains as much variation as more than six original GI variables.

Interpretation. The factor is not chosen because it predicts returns; it is chosen only because it summarizes the dominant dimension of time variation in the column's content.

2.5 What the factor means in plain English

The first factor loads positively on only a handful of categories, most importantly:

- **Negative**,
- **Weak**,
- **Fail**,
- **Fall**.

Negative and Weak each explain more than **57%** of the variance in the factor, and the factor loads negatively on positive words.

Interpretation. High values of the factor correspond to a column that reads as pessimistic, weak, or failure-oriented. That is why the factor is labeled **pessimism**.

2.6 Main outcome variables

1. Dow Jones returns. The paper’s main return outcome is the daily return on the Dow Jones Industrial Average, measured in basis points.

2. SMB factor. To test whether sentiment matters more for small stocks, the paper studies the daily Fama-French **small-minus-big** factor.

3. NYSE trading volume. Volume is measured as detrended log NYSE trading volume. If $\log Vol_t$ is log daily volume, the detrended series is conceptually

$$Vlm_t = \log Vol_t - \frac{1}{60} \sum_{s=1}^{60} \log Vol_{t-s}. \quad (2)$$

The paper uses a 60-day rolling mean and shows robustness to 30-day and 360-day windows.

Interpretation. Volume is used to test whether pessimism proxies for disagreement, liquidity trading, or trading costs.

2.7 Controls and timing variables

The baseline predictive system includes:

- five lags of returns,
- five lags of detrended log volume,
- five lags of detrended squared Dow residuals as a proxy for past volatility,
- day-of-the-week dummies,
- a January dummy,
- and a dummy for the October 19, 1987 stock market crash.

Interpretation. The regressions are designed to ask whether media pessimism predicts *additional* market variation beyond common short-horizon return and volume dynamics.

2.8 Alternative sentiment proxies

Besides the principal-component factor, the paper uses two simpler GI-based measures:

- **Negative** words,
- **Weak** words.

These are important because they are psychologically interpretable and help check that the results are not an artifact of the PCA construction.

2.9 Descriptive facts worth remembering

- Unconditional mean Dow Jones daily return over the sample is about **5.4 basis points**.
- One standard deviation increase in pessimism is therefore large relative to the mean daily return.
- The semiparametric fit later in the paper shows roughly a **25 basis point** return gap between days with very low versus very high sentiment (bottom 5% versus top 5%).

Interpretation. The measured sentiment effects are economically nontrivial relative to daily return magnitudes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper tests three broad interpretations of media tone.

1. Sentiment / noise-trader mechanism. Following models such as DeLong et al. (1990a), high pessimism means noise traders become more pessimistic than rational arbitrageurs. Then:

- they sell stocks,
- prices fall temporarily,
- returns later reverse when sentiment mean-reverts,
- and unusually high or low sentiment increases volume.

2. Liquidity / risk-aversion mechanism. Following Campbell, Grossman, and Wang (1993), large shifts in demand by a subset of investors can temporarily depress returns and raise volume.

3. Information mechanism. If pessimism is a proxy for genuine negative news about cash flows, then returns should fall and *stay* low rather than reverse.

Interpretation. The sign pattern over horizons—immediate decline plus later reversal—is central to the paper’s argument.

3.2 Baseline return equation for the Dow

Let $L_5(x_t) = [x_{t-1}, x_{t-2}, x_{t-3}, x_{t-4}, x_{t-5}]$ denote the vector of five lags of variable x_t . The main return equation is

$$Dow_t = \alpha_1 + \beta_1 \cdot L_5(Dow_t) + \gamma_1 \cdot L_5(BdNws_t) + \delta_1 \cdot L_5(Vlm_t) + \lambda_1 \cdot Exog_{t-1} + \varepsilon_{1t}. \quad (3)$$

Here:

- Dow_t is the daily Dow return,
- $BdNws_t$ is the pessimism measure,
- Vlm_t is detrended log NYSE volume,
- $Exog_{t-1}$ collects volatility and calendar controls.

Interpretation. The coefficients on $L_5(BdNws_t)$ ask whether negative media tone forecasts future returns after controlling for recent market behavior.

3.3 Media-feedback equation

To see whether the market also predicts future media tone, the paper estimates

$$BdNws_t = \alpha_2 + \beta_2 \cdot L_5(Dow_t) + \gamma_2 \cdot L_5(BdNws_t) + \delta_2 \cdot L_5(Vlm_t) + \lambda_2 \cdot Exog_{t-1} + \varepsilon_{2t}. \quad (4)$$

Interpretation. This is not a structural causal model. It simply checks whether negative returns are followed by more pessimistic media coverage.

3.4 Volume equation

The volume regression adds both the level and the absolute value of pessimism:

$$Vlm_t = \alpha_3 + \beta_3 \cdot L_5(Dow_t) + \gamma_3 \cdot L_5(BdNws_t) + \psi_3 \cdot L_5(|BdNws_t|) + \delta_3 \cdot L_5(Vlm_t) + \lambda_3 \cdot Exog_{t-1} + \varepsilon_{3t}. \quad (5)$$

Interpretation.

- γ_3 tests whether signed pessimism directly shifts volume.
- ψ_3 tests whether extreme sentiment—positive or negative relative to the mean—raises volume through disagreement or noninformational trading.

3.5 Small-stock return equation

The paper next examines whether sentiment matters especially for small stocks:

$$SMB_t = \alpha_4 + \beta_4 \cdot L_5(Dow_t) + \gamma_4 \cdot L_5(BdNws_t) + \delta_4 \cdot L_5(Vlm_t) + \pi_4 \cdot L_5(SMB_t) + \lambda_4 \cdot Exog_{t-1} + \varepsilon_{4t}. \quad (6)$$

Interpretation. If sentiment is tied more closely to individual investors, it should matter more for assets held heavily by those investors. Small stocks fit that idea better than the Dow.

3.6 Estimation approach

- Each equation is estimated by **OLS**.
- The paper treats the system as a reduced-form VAR / Granger-causality setup.
- Newey-West standard errors allow for heteroskedasticity and autocorrelation up to five lags.

Interpretation. The paper is not trying to identify a deep structural model of price formation. It is testing whether lagged media content predicts market outcomes in a disciplined time-series environment.

3.7 Robust timing specifications

Because the column is released near the market close and then reprinted next morning, the paper asks whether results merely reflect slow incorporation of hard news.

It therefore re-estimates the return regression with a more conservative dependent variable:

$$Dow_t^{10am-close}, \tag{7}$$

namely the Dow return from **10 a.m. to the market close** the day after the article is released.

Similarly, it defines an **after-morning volume** outcome by removing opening-hour volume from daily NYSE volume.

Finally, for SMB robustness, it shifts the sentiment window back by one additional day and predicts SMB with **lags 2 through 6** of sentiment rather than lags 1 through 5.

Interpretation. These checks are designed to ask whether the predictive content of pessimism survives after traders have had substantial time to absorb any literal news in the article.

3.8 Semiparametric specification

To check nonlinearities, the paper first residualizes Dow returns using all controls from equation (3) except the lags of sentiment, then fits locally weighted regressions of those residual returns on sentiment.

Interpretation. This lets the paper verify whether the return-sentiment relation is monotone and whether it is driven only by a few outliers.

3.9 Economic-importance benchmark: trading strategy

The paper also constructs a simple benchmark strategy using the **Negative** word measure:

- if Negative words are in the bottom third of the prior year's distribution, go **long** the Dow for one day;
- if Negative words are in the top third, go **short** the Dow for one day.

Interpretation. This is not presented as a realistic implementable strategy. It is an economic-magnitude benchmark for the predictive content of the media signal.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline estimate

The return prediction result comes from daily time-series variation in media pessimism after controlling for:

- lagged returns,
- lagged volume,
- lagged volatility,
- weekday effects,
- January effects,
- and the 1987 crash.

Interpretation. The identifying comparison is not simply “bad news days” versus “good news days.” It is days with unusually pessimistic media tone relative to the recent behavior of the market itself.

4.2 Why reverse causality is only part of the story

The paper openly finds that low returns predict higher next-day pessimism. That is, the media partly *reacts* to the market. But the core result is the converse direction: pessimism also predicts future returns.

Interpretation. The evidence supports a feedback loop rather than a one-way story. Markets move the media, and media tone forecasts subsequent market activity.

4.3 Why slow incorporation of hard news is unlikely to explain the results

A simple alternative story is: the article contains fundamental information released near the close, and prices incorporate it only the next morning.

The paper pushes against this interpretation in three ways:

- the article is written around the close and often before the close, so much of any hard information is already public;

- sentiment still predicts returns from **10 a.m. to close** the next day;
- the effect is spread through the following week, not concentrated only right after release.

Interpretation. If this were simply delayed reaction to news, the effect should be strongest immediately after release and should not require days to unwind.

4.4 Why the fundamental-information view is weak

The fundamental-information hypothesis predicts continuation, not reversal.

What the paper finds instead:

- returns fall immediately after pessimistic tone,
- then they reverse over the next several days,
- and the sum of coefficients is often near zero for the Dow.

Interpretation. That pattern looks like temporary price pressure, not updated cash-flow fundamentals.

4.5 Why the stale-information / sideshow view is weak

If the article merely recycles information already in prices, sentiment should not forecast future returns or future volume.

But the paper finds:

- strong return predictability,
- strong volume predictability from absolute pessimism,
- and similar patterns for multiple sentiment proxies.

Interpretation. The column may still contain entertainment or commentary, but it is not an irrelevant sideshow statistically.

4.6 Risk-premium explanation and why it fails

Another alternative is that pessimism predicts lower expected returns because risk is temporarily lower on pessimistic days.

The paper checks this and finds little support:

- it cannot find convincing evidence that pessimism proxies for lower risk;
- if anything, pessimism weakly predicts **higher** volatility.

Interpretation. The risk-premium channel points the wrong way.

4.7 Alternative proxies and data-mining concerns

The results replicate when the pessimism factor is replaced by the simpler GI categories **Negative** and **Weak**. These variables were designed by psychologists, not selected because they maximize return predictability.

Interpretation. This helps credibility in two ways: the results are more interpretable, and they are less likely to be a PCA artifact or data-mining accident.

4.8 What the paper can and cannot distinguish

The paper is careful about limits.

- It **can** reject the view that media tone is just stale information or purely fundamental news.
- It **cannot** cleanly distinguish between investor sentiment, risk-aversion shocks, and liquidity-demand shocks.
- It also cannot fully separate whether the media *causes*, *amplifies*, or merely *reflects* sentiment.

Interpretation. The safe takeaway is that media tone behaves like a real-time sentiment proxy, not that the paper identifies a single psychological mechanism with certainty.

4.9 Relation to the literature

- Relative to **Cutler, Poterba, and Summers (1989)**, the paper shows that qualitative media content can forecast broad market activity.
- Relative to **Antweiler and Frank (2004)** and **Coval and Shumway (2001)**, it links communication measures not just to volume but also to returns.
- Relative to **DeLong et al. (1990a)** and **Campbell, Grossman, and Wang (1993)**, it provides market-wide evidence that looks like temporary sentiment or liquidity pressure.

5 Tables and figures

5.1 Figure 1: theory of the sentiment shock

Read. Figure 1 gives the paper's conceptual roadmap. If pessimism predicts low future sentiment, it should forecast low short-horizon returns and then high long-horizon returns. If pessimism mainly reflects past low sentiment, then the long-horizon rebound should be even stronger than the short-horizon drop.

Economic message. The reversal pattern is what separates sentiment from fundamental news.

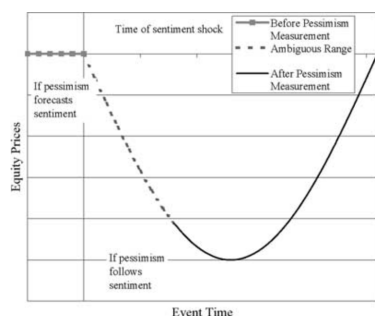


Figure 1. Impact of a negative sentiment shock on prices. The graph depicts the theoretical impact of a one-time increase in negative investor sentiment on equity prices. If the media pessimism measure is a predictor of investor sentiment, it will predict low short-horizon returns followed by high long-horizon returns of approximately equal magnitude. If the media pessimism measure follows past investor sentiment, it will predict low short-horizon returns followed by high long-horizon returns of greater magnitude than the short-horizon returns.

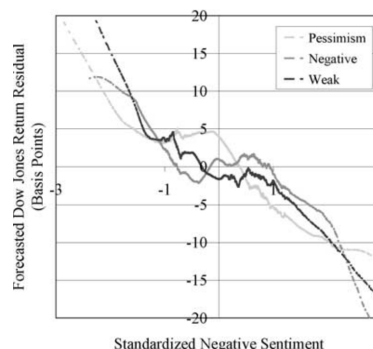


Figure 2. The effect of negative sentiment on Dow Jones returns. Data from the author's application of the General Inquirer program to the *Wall Street Journal* column from January 2, 1984 to September 17, 1999. The Dow-Jones residuals come from the regression depicted in equation (1). The three measures of negative sentiment are calculated using the factor analysis procedure described in Section III. In the original unsmoothed data, there are 3,709 observations on daily residual Dow Jones returns and negative sentiment. Each smoothed data point represents the fitted value from a locally weighted least squares regression of residual Dow Jones returns on negative sentiment. In each local regression, I apply local weightings from the standard tricube weighting function to only neighboring values of negative sentiment. The neighborhood for each data point is centered around the data point and includes one-half of the sample.

5.2 Table I: the factor is stable across years

Read. Table I shows that yearly pessimism factors built from different years' loadings are highly correlated with one another. Average pairwise correlation is about **0.96**; the minimum is about **0.80**.

Economic message. The factor is not a fragile accident of one year or one writer. It captures a persistent semantic dimension of the column.

5.3 Table II: the baseline return effect

Read.

- One-standard-deviation increase in **Pessimism**: next-day Dow return falls by about **8.1 bps**.
- Days 2–5 reversal sums to about **6.8 bps**.
- **Negative** words: next-day effect about **-4.4 bps**, reversal about **+9.5 bps**.
- **Weak** words: next-day effect about **-6.0 bps**, reversal about **+10.7 bps**.

The joint hypothesis that pessimism does not forecast Dow returns is strongly rejected.

Economic message. Media pessimism predicts a temporary decline in prices, not a permanent repricing.

5.4 Table III: returns predict future pessimism

Read. A 1% decline in the Dow predicts an increase in pessimism of about **5.8% of one standard deviation** the next day. Similar patterns hold for Negative and Weak word counts.

Economic message. The media partly follows the market. This makes the environment one of feedback rather than one-way causation.

Table II Predicting Dow Jones Returns Using Negative Sentiment			
The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_1 in equation (1). Each coefficient measures the impact of a one-standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). The regression is based on 3,709 observations from January 1, 1984, to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.			
News Measure	Regressand: Dow Jones Returns		
	Pessimism	Negative	Weak
$BdNws_{t-1}$	-8.1	-4.4	-6.0
$BdNws_{t-2}$	0.4	3.6	2.0
$BdNws_{t-3}$	0.5	-2.4	-1.2
$BdNws_{t-4}$	4.7	4.4	6.3
$BdNws_{t-5}$	1.2	2.9	3.6
$\chi^2(5)$ [Joint]	20.0	20.8	26.5
p-value	0.001	0.001	0.000
Sum of 2 to 5	6.8	9.5	10.7
$\chi^2(1)$ [Reversal]	4.05	8.35	10.1
p-value	0.044	0.004	0.002

Table III Predicting Negative Sentiment Using Dow Jones Returns			
The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient β_2 in equation (2). Each coefficients measures the effect of a 1% increase in Dow Jones returns on negative investor sentiment in units of standard deviations. The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.			
Regressand	Pessimism	Negative	Weak
Dow_{t-1}	-5.8	-7.6	-5.6
Dow_{t-2}	2.3	-0.7	0.7
Dow_{t-3}	2.1	1.2	1.9
Dow_{t-4}	2.3	0.3	1.7
Dow_{t-5}	4.2	1.5	1.0
$\chi^2(5)$ [Joint]	16.6	14.1	12.1
p-value	0.006	0.015	0.033

5.5 Table IV: signed pessimism and absolute pessimism in volume

Read.

- Signed pessimism itself is at best a weak and unstable direct predictor of volume.
- But the **absolute value** of pessimism strongly predicts next-day volume increases.
- A one-standard-deviation increase in absolute pessimism raises detrended log volume by roughly **1.3% to 1.5%** on the next day.

Economic message. Extreme tone—whether unusually negative or unusually nonnegative relative to normal—looks like a proxy for disagreement or unusual nonfundamental order flow.

Table IV Predicting Log NYSE Volume Using Sentiment			
The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_3 in equation (3). The two sets of coefficients describe the impact of one-standard deviation increases in negative sentiment and the absolute value of sentiment on detrended log NYSE volume growth. The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.			
News Measure	Regressand: Volume		
	Pessimism	Negative	Weak
$BdNws_{t-1}$	-0.80	-0.18	-0.80
$BdNws_{t-2}$	-0.16	-0.16	0.29
$BdNws_{t-3}$	0.31	0.04	0.68
$BdNws_{t-4}$	-0.11	0.33	0.87
$BdNws_{t-5}$	0.18	0.13	0.15
$ BdNws_{t-1} $	1.34	1.51	1.34
$ BdNws_{t-2} $	-1.50	-0.37	-0.34
$ BdNws_{t-3} $	-0.10	0.38	-0.30
$ BdNws_{t-4} $	0.71	0.41	0.60
$ BdNws_{t-5} $	-1.97	0.15	-0.62
$\chi^2(5)$ [BdNws]	6.6	2.2	15.4
p-value	0.256	0.823	0.009
$\chi^2(5)$ [BdNws]	34.9	13.2	10.7
p-value	0.000	0.022	0.058

Table V Predicting the Small-Minus-Big Factor Using Negative Sentiment			
The table data come from CRSP, NYSE, Professor Kenneth French's web site, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_4 in equation (4). Each coefficient measures the impact of a one-standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.			
News Measure	Regressand: Small-Minus-Big Returns		
	Pessimism	Negative	Weak
$BdNws_{t-1}$	-2.0	-1.2	-0.9
$BdNws_{t-2}$	1.1	-0.5	-0.6
$BdNws_{t-3}$	-0.4	0.1	0.3
$BdNws_{t-4}$	-2.4	-2.4	-2.7
$BdNws_{t-5}$	-0.9	-1.6	-2.0
$\chi^2(5)$ [Joint]	12.1	15.5	13.7
p-value	0.034	0.008	0.018
Sum 1 to 5	-4.6	-5.6	-5.9
$\chi^2(1)$ [Sum]	5.5	11.7	9.5
p-value	0.019	0.001	0.020

5.6 Tables V to VIII: small stocks and timing robustness

Read.

- Table V shows pessimism predicts negative **SMB** returns over the following week: cumulative effects are around **-4.6 to -5.9 bps**.

Table VI
Robustness of Dow Jones Return Forecasts Using Negative Sentiment

The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_1 in equation (1). The return time window for the dependent variable in equation (1) has been shortened to include only the returns from 10 a.m. to market closing time on the day after the news column is released. Each coefficient measures the impact of a one-standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.

News Measure	Regressand: Dow Jones Returns		
	Pessimism	Negative	Weak
$BdNews_{t-1}$	-6.2	-3.7	-4.5
$BdNews_{t-2}$	0.9	1.9	1.2
$BdNews_{t-3}$	-0.9	-2.8	-2.2
$BdNews_{t-4}$	4.0	4.0	5.3
$BdNews_{t-5}$	4.6	2.5	3.0
$\chi^2(5)$ [Joint]	17.5	21.5	24.0
p-value	0.004	0.001	0.000
Sum of 2 to 5	5.4	5.6	7.3
$\chi^2(1)$ [Reversal]	3.6	5.4	6.8
p-value	0.058	0.021	0.009

Table VII
Robustness of Log NYSE Volume Forecasts Using Sentiment

The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_1 in equation (2). The definition of volume in equation (2) has been modified to include only trades occurring between 10 a.m. and market closing time on the day after the news column is released. The two sets of coefficients describe the impact of one-standard deviation increases in negative sentiment and the absolute value of sentiment on detrended log NYSE volume growth. The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.

News Measure	Regressand: Volume		
	Pessimism	Negative	Weak
$BdNews_{t-1}$	-0.52	0.01	-0.49
$BdNews_{t-2}$	-0.07	-0.09	0.22
$BdNews_{t-3}$	0.43	-0.49	0.88
$BdNews_{t-4}$	-0.12	0.42	1.01
$BdNews_{t-5}$	0.08	0.38	0.14
$[BdNews_{t-1}]$	1.17	1.51	1.34
$[BdNews_{t-2}]$	1.52	-0.43	-0.16
$[BdNews_{t-3}]$	0.01	0.42	-0.33
$[BdNews_{t-4}]$	1.02	0.51	0.42
$[BdNews_{t-5}]$	-1.88	0.13	-0.63
$\chi^2(5)$ [Joint]	3.2	2.0	14.1
p-value	0.676	0.852	0.015
$\chi^2(1)$ [BdNews]	21.8	12.0	8.4
p-value	0.000	0.005	0.137

Table VIII
Robustness of SMB Return Forecasts Using Negative Sentiment

The table data come from CRSP, NYSE, Professor Kenneth French's web site, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_2 in equation (4); however, equation (4) has been modified to include the second through sixth lags of the pessimism factor rather than the first through fifth lags. Each coefficient measures the impact of a one-standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). The regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; italics and bold denotes significance at the 1% level.

News Measure	Regressand: Small-Minus-Big Returns		
	Pessimism	Negative	Weak
$BdNews_{t-2}$	0.2	-1.1	-1.2
$BdNews_{t-3}$	-0.3	-0.0	-0.0
$BdNews_{t-4}$	-2.8	-2.7	-3.0
$BdNews_{t-5}$	-1.3	-2.0	-2.3
$BdNews_{t-6}$	-2.0	-1.0	-1.6
$\chi^2(5)$ [Joint]	14.3	16.6	16.6
p-value	0.014	0.005	0.005
Sum 2 to 6	-6.7	-6.8	-6.0
$\chi^2(1)$ [Joint]	9.7	14.8	13.8
p-value	0.002	0.000	0.000

- Table VI shows that even when the return window begins at **10 a.m.** the next day, pessimism still predicts a Dow decline of about **-6.2 bps** with later reversal.
- Table VII shows that **after-morning volume** still responds strongly to the absolute value of pessimism.
- Table VIII shows SMB predictability survives when sentiment is shifted back another day (lags 2–6), with cumulative effects around **-6.7 to -8.0 bps**.

Economic message. The predictive content of media tone is not just an opening-bell effect. It lasts long enough to look like sentiment-driven price pressure rather than slow reaction to literal news.

5.7 Table IX and Figure 2: subperiods and nonlinearities

Table IX
Predicting Dow Jones Returns Using Negative Sentiment

The table data come from CRSP, NYSE, and the General Inquirer program. This table shows OLS estimates of the coefficient γ_1 in equation (1). Each coefficient measures the impact of a one-standard deviation increase in negative investor sentiment on returns in basis points (one basis point equals a daily return of 0.01%). Each regression is based on 3,709 observations from January 1, 1984 to September 17, 1999. I use Newey and West (1987) standard errors that are robust to heteroskedasticity and autocorrelation up to five lags. Bold denotes significance at the 5% level; and italics and bold denotes significance at the 1% level.

News Measure	Regressand: Dow Jones Returns					
	1984–1991			1992–1999		
	Pessimism	Negative	Weak	Pessimism	Negative	Weak
$BdNews_{t-1}$	-0.5	3.1	-0.1	-13.6	-10.9	-9.8
$BdNews_{t-2}$	-3.7	0.7	-1.2	3.3	5.8	3.3
$BdNews_{t-3}$	-2.3	-3.1	-2.8	2.4	-2.3	-0.5
$BdNews_{t-4}$	6.7	5.0	9.3	3.1	3.4	4.6
$BdNews_{t-5}$	-2.8	2.6	2.7	2.9	2.6	3.8
$\chi^2(5)$ [Joint]	5.9	9.3	9.3	42.2	31.1	28.3
p-value	0.316	0.099	0.098	0.000	0.000	0.000
Sum of 2 to 5	-2.1	5.2	8.0	11.7	9.5	11.3
$\chi^2(1)$ [Reversal]	0.2	1.4	1.8	6.7	10.7	7.2
p-value	0.681	0.234	0.183	0.010	0.018	0.007

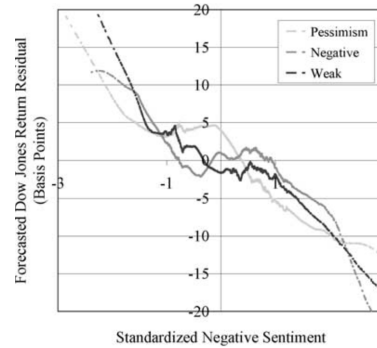


Figure 2. The effect of negative sentiment on Dow Jones returns. Data from the author's application of the General Inquirer program to the *Wall Street Journal* column from January 2, 1984 to September 17, 1999. The Dow Jones residuals come from the regression depicted in equation (1). The three measures of negative sentiment are calculated using the factor analysis procedure described in Section III. In the original unsmoothed data, there are 3,709 observations on daily residual Dow Jones returns and negative sentiment. Each smoothed data point represents the fitted value from a locally weighted least squares regression of residual Dow Jones returns on negative sentiment. In each local regression, I apply local weightings from the standard tricube weighting function to only neighboring values of negative sentiment. The neighborhood for each data point is centered around the data point and includes one-half of the sample.

Read.

- Table IX shows that effects are much stronger in **1992–1999** than in **1984–1991**. In the later period, the next-day pessimism effect on the Dow is about **-13.6 bps** for the factor and around **-10 bps** for Negative and Weak words.

- Figure 2 shows a monotone downward relation between sentiment and residual Dow returns in a lowess fit.
- The gap in expected returns between very low sentiment days and very high sentiment days (bottom 5% versus top 5%) is about **25 bps**.

Economic message. Sentiment mattered more in the 1990s and the effect is not driven by one or two extreme outliers. The nonlinearity mainly says the effect gets stronger in the tails.

5.8 Economic importance and the trading-strategy benchmark

Read.

- The five lags of the pessimism factor explain about **1.52%** of residual Dow-return variation.
- That may sound small, but it is larger than the explanatory contribution of lagged returns, lagged volume, or calendar dummies in this setting.
- A simple benchmark trading rule based on **Negative** words yields about **4.4 bps** per day, or about **7.3% annualized**, before transaction costs.
- The strategy has positive yearly expected returns in **12 of 15 years**.

Economic message. The signal is not just statistically significant; it is economically meaningful. But it is probably too fragile after trading costs, bid-ask spreads, market depth, and taxes.

6 Short summary

- The paper turns daily WSJ text into a pessimism measure using quantitative content analysis.
- More pessimistic media tone predicts lower next-day Dow returns and a later reversal.
- Extreme sentiment predicts higher trading volume.
- The same patterns appear for simpler word categories (Negative and Weak), which improves interpretability.
- Small-stock returns react more persistently than the Dow, consistent with a stronger role for individual-investor sentiment.
- The evidence is hard to reconcile with a pure information story or a stale-news story.
- The paper's safest interpretation is that media content is a useful proxy for investor sentiment or noninformational trading pressure.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that daily financial-media tone—measured from the content of a major Wall Street Journal column—predicts broad stock market returns and trading volume. High pessimism forecasts a short-run fall in prices, followed by reversal, and extreme sentiment forecasts higher volume.

7.2 Economic intuition

The simplest intuition is temporary price pressure. When tone is unusually pessimistic, some investors become more negative or more risk averse than others, sell into the market, and push prices down. Arbitrage eventually undoes the mispricing, so returns later rebound. The same disagreement or liquidity pressure raises trading volume when sentiment is unusually far from normal.

7.3 Takeaway

This paper is important because it shows that *what the media says*—not just the arrival of obvious news events—contains information about investor sentiment. It provides a clean early example of text analysis in finance and a micro-style bridge between behavioral finance theories and observable market-wide data.